



QUALITY IN SPORT

eISSN 2450-3118 · Open Access · Peer-reviewed

apcz.umk.pl/QS Nicolaus Copernicus University in Toruń



KLIMAS, Paweł, WITASZCZYK, Małgorzata, SOLTAN, Alicja, WIKTORZAK, Natalia, LIPSKA, Julia, PRZĄDKA, Jagoda, KOCHAN-OLSZEWSKA, Dominika, RUDNICKI, Longin, PAWLICKI, Jakub, KOŻMA, Gracjan, LASKUS, Justyna, GAWIOR, Bartosz, STASIAK, Olga and BORKOWSKA, Marta. Sleep quantity and quality as potentially modifiable factors associated with musculoskeletal injury risk in athletes: a narrative review. *Quality in Sport*. 2026;72:74894. <https://doi.org/10.12775/QS.2026.72.74894>

ARTICLE TIMELINE

Received: 02.08.2026. Revised: 21.08.2026. Accepted: 07.09.2026. Published: 15.09.2026.

The journal has been awarded 20 points in the parametric evaluation by the Polish Ministry of Higher Education and Science (Annex to the announcement of 05.01.2024, No. 32553). Unique Journal Identifier: 201398. Scientific disciplines assigned: Economics and Finance (Field of Social Sciences); Management and Quality Sciences (Field of Social Sciences).

Punkty Ministerialne z 2019 – aktualny rok 20 punktów. Załącznik do komunikatu Ministra Szkolnictwa Wyższego i Nauki z dnia 05.01.2024 Lp. 32553. Posiada Unikatowy Identyfikator Czasopisma: 201398. Przypisane dyscypliny naukowe: Ekonomia i finanse (Dziedzina nauk społecznych); Nauki o zarządzaniu i jakości (Dziedzina nauk społecznych). © The Authors 2026.

OPEN ACCESS · CC BY-NC-SA 4.0 This article is published with open access under the License Open Journal Systems of Nicolaus Copernicus University in Toruń, Poland, and is distributed under the terms of the Creative Commons Attribution Non-commercial Share Alike License (<http://creativecommons.org/licenses/by-nc-sa/4.0/>), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the work is properly cited. The authors declare no conflict of interest regarding the publication of this paper.

Sleep quantity and quality as potentially modifiable factors associated with musculoskeletal injury risk in athletes: a narrative review

Paweł Klimas, ORCID <https://orcid.org/0009-0002-8803-0476>

- E-mail klimas.pawel42@gmail.com

Św. Wincentego a Paulo Hospital, 1 Wójta Radtkego St., 81-348 Gdynia, Poland

Małgorzata Witaszczyk, ORCID <https://orcid.org/0009-0004-0197-5480>

- E-mail malgosiawitaszczyk@gmail.com

University Clinical Center of the Medical University of Silesia in Katowice, ul. Medyków 18, 40-752 Katowice, Poland

Alicja Sołtan, ORCID <https://orcid.org/0009-0006-8750-0797>

- E-mail soltan.alicja@gmail.com

Central Clinical Hospital of the Medical University of Warsaw, ul. Banacha 1A, 02-097 Warsaw, Poland

Natalia Wiktorzak, ORCID <https://orcid.org/0009-0004-8601-3087>

- E-mail natalia.wiktorzak@onet.pl

Cardinal Stefan Wyszyński Voivodeship Hospital in Łomża, Marszałka Józefa Piłsudskiego 11, 18-400 Łomża, Poland

Julia Lipska, ORCID <https://orcid.org/0009-0009-4093-2543>

- E-mail Julka98712@gmail.com

Medical University of Lublin, al. Raławickie 1, 20-059 Lublin, Poland

Jagoda Prządka, ORCID <https://orcid.org/0009-0001-7921-919X>

- E-mail jagodaprzadka45@gmail.com

Nicolaus Copernicus University in Toruń, Ludwik Rydygier Collegium Medicum in Bydgoszcz, ul. Jagiellońska 15, 85-067 Bydgoszcz, Poland

Dominika Kochan-Olszewska, ORCID <https://orcid.org/0009-0002-0001-7809>

- E-mail dominikakochan98@gmail.com

Holycross Cancer Center in Kielce, ul. Artwińskiego 3, 25-734 Kielce, Poland

Longin Rudnicki, ORCID <https://orcid.org/0009-0003-8980-8718>

- E-mail longinr@op.pl

Gdańsk Medical University, Gdańsk, Poland

Jakub Pawlicki, ORCID <https://orcid.org/0009-0007-3888-4672>

- E-mail jakub.pawlicki96@gmail.com

Specialist Hospital St. Wojciech, Aleja Jana Pawła II 50, 80-462 Gdańsk, Poland

Gracjan Koźma, ORCID <https://orcid.org/0009-0002-8812-6427>

- E-mail gracjan.kozma@gmail.com

Military Institute of Aviation Medicine, 54/56 Krasińskiego Street, 01-755 Warsaw, Poland

Justyna Laskus, ORCID <https://orcid.org/0009-0004-8065-0464>

- E-mail jlaskus@poczta.onet.pl

Bieleński Hospital, ul. Ceglowska 80, 01-809 Warsaw, Poland

Bartosz Gawior, ORCID <https://orcid.org/0009-0003-0935-5414>

- E-mail bartosz.gawior@gmail.com

Mazovian Bródno Hospital, ul. Ludwika Kondratowicza 8, 03-242 Warsaw, Poland

Olga Stasiak, ORCID <https://orcid.org/0009-0008-3013-8015>

- E-mail stasiak.olga1@gmail.com

Międzylesie Specialist Hospital in Warsaw, ul. Bursztynowa 2, 04-749 Warsaw, Poland

Marta Borkowska, ORCID <https://orcid.org/0009-0002-8105-3384>

- E-mail marborkk@gmail.com

Military Institute of Medicine – National Research Institute, ul. Szaserów 128, 04-141 Warsaw, Poland

Corresponding Author

Paweł Klimas, E-mail: klimas.pawel42@gmail.com

Abstract

Background. Sleep is increasingly seen as a potentially modifiable factor in athlete health, but its link to injury risk is not well established across ages and sports.

Aim. To summarize evidence on sleep quantity and quality as potential risk factors for musculoskeletal injury in athletes and identify areas of consistency and uncertainty.

Material and methods. A narrative review was conducted, based on a structured but incompletely documented PubMed search supplemented by Consensus-assisted literature identification and informed by SANRA.

Results. In adolescents, cross-sectional and prospective studies, with a broader review of adolescent sleep and injury literature, reported associations between short or poor sleep and injury risk, with some overlap in cohorts and non-athlete populations. In adults, evidence is heterogeneous, and findings from self-reported and actigraphy-based sleep measures have not been uniform; a systematic review did not confirm sleep as an independent risk factor. Sex, sport, and training load may relate to differences in risk, although evidence of effect modification remains unconfirmed.

Conclusions. Short sleep, and less consistently poor sleep quality, are associated with injury risk, particularly in adolescents, and may represent potentially modifiable factors; however, causality and preventive efficacy have not been established, and evidence in adults remains inconsistent. Prospective studies using reproducible methods are needed.

Key words: Sleep; Sleep Duration; Sleep Quality; Athletes; Athletic Injuries; Musculoskeletal Injuries; Injury Prevention; Adolescent.

1. INTRODUCTION AND PURPOSE	5
2. MATERIAL AND METHODS	6
2.1. Review design	6
2.2. Literature search	6
2.3. Eligibility criteria	7
2.4. Study selection and evidence use	7
2.5. Methodological limitations of this review	7
3. REVIEW OF AVAILABLE EVIDENCE	8
3.1. Definitions and measurement of sleep quantity and quality	8
3.2. Sleep characteristics in athletic populations	8
3.3. Sleep duration and injury risk	9
3.4. Sleep quality and injury risk	10
3.5. Evidence in youth athletes	11
3.6. Evidence in adult, collegiate, and elite athletes	12
3.7. Differences by sex, sport, and training context	13
3.8. Potential biological and behavioral mechanisms	13
3.9. Sleep interventions and injury prevention	14
3.10. Limitations and inconsistencies of the evidence	15
4. DISCUSSION	19
5. CONCLUSIONS	21
Disclosure	22
Supplementary Materials	22
Author Contributions	22
Funding	22
Institutional Review Board Statement	22
Informed Consent Statement	22
Data Availability Statement	22
Acknowledgments	22
Conflicts of Interest	22
Originality	22
Declaration of the use of generative AI and AI-assisted technologies in the writing proces	23
References	23

1. INTRODUCTION AND PURPOSE

Musculoskeletal injury remains one of the most consequential threats to athlete health, sport participation, and career longevity. Time-loss and medical-attention injuries interrupt training continuity, delay long-term athletic development, and in youth populations carry the additional risk of premature disengagement from sport. Identifying modifiable risk factors for injury is therefore a recurring priority in sports medicine research, alongside training load, biomechanics, and psychosocial stress.

Sleep has emerged as one such candidate factor. Sleep supports tissue repair, hormonal regulation, immune function, and neurocognitive processes that are directly relevant to injury risk, including reaction time, motor coordination, and attentional control ¹. Athletic populations are, however, particularly prone to inadequate sleep. Congested competition schedules, early-morning training, travel across time zones, and psychological stress before competition have all been described as contributing to shortened or fragmented sleep in athletes ^{1,2,3}. Narrative and mechanistic reviews have proposed that this pattern of sleep curtailment could translate into a higher likelihood of musculoskeletal injury through impaired neuromuscular control, delayed reaction times, and altered anabolic-catabolic hormonal balance ^{4,5}.

Despite this plausible mechanistic basis, empirical support for sleep as a risk factor for musculoskeletal injury in athletes is not uniform. Several studies in adolescent athletes have reported that habitually short sleep duration is associated with a higher likelihood of sustaining a sports injury ^{6,7,8}. In adult and elite athletic populations, however, the picture is considerably less consistent. A systematic review restricted to adult athletes concluded that current evidence does not support poor sleep as an independent risk factor for sport-related injury, in contrast with the adolescent literature ⁹. Individual studies using objective, actigraphy-based sleep measurement have likewise failed to confirm an association in elite team-sport athletes ¹⁰, while other prospective cohorts using self-reported sleep quality or duration have reported clear associations with subsequent injury ^{11,12,13,14}. This divergence may reflect differences in study design, in how sleep is measured (self-report versus actigraphy), in how injury is defined (time-loss versus medical-attention versus any physical complaint), in the length and intensity of follow-up, and in the developmental stage, sex, and sport of the athletes studied.

A further source of uncertainty concerns the distinction between short sleep duration, poor subjective sleep quality, and clinically diagnosed sleep disorders such as chronic insomnia. These exposures are conceptually related but not interchangeable, and studies in athletic and non-athletic populations have not always distinguished between them ¹⁵. Evidence from non-athletic and military populations, where newly diagnosed insomnia identified from administrative claims and poor sleep quality have both been linked to increased musculoskeletal injury risk, provides useful context but cannot be assumed to generalize directly to competitive athletes, whose injury mechanisms, training exposures, and baseline health status differ substantially ^{16,17,18}.

Given this heterogeneity, a systematic quantitative synthesis was judged premature, and a narrative review was considered a more appropriate approach for mapping the current state of

evidence, highlighting areas of convergence and divergence, and identifying the methodological features that most plausibly explain inconsistent findings across studies. Narrative synthesis also allows explicit discussion of studies that are informative but not directly comparable, such as mechanistic reports, intervention studies targeting sleep for performance rather than injury outcomes, and evidence from non-athletic but physically active populations.

The aim of this narrative review was to summarize and critically appraise the available evidence on sleep quantity and sleep quality as potential risk factors for musculoskeletal injury in athletes, across adolescent, collegiate, and adult populations, and to identify the principal sources of inconsistency, the plausible underlying mechanisms, and the gaps that should guide future prospective research.

2. MATERIAL AND METHODS

2.1. Review design

This is a narrative review, not a systematic review or meta-analysis. It was prepared with reference to the Scale for the Assessment of Narrative Review Articles (SANRA), which addresses the justification of the review's importance, a clearly stated aim, a description of the literature search, appropriate and representative citation of evidence, sound scientific reasoning, and appropriate presentation of the relevant data on study outcomes¹⁹.

2.2. Literature search

A structured narrative literature search was conducted in PubMed, with the final search performed on 28 July 2026. PubMed was the sole bibliographic database; Consensus was used only as a supplementary literature-identification tool. The search combined terms related to sleep (“sleep duration”, “sleep quality”, “sleep disturbance”, “sleep deprivation”, “insomnia”, and “actigraphy”) with terms related to athletic populations (“athlete”, “sport”, “player”, and “runner”) and injury outcomes (“sports injury”, “athletic injury”, and “musculoskeletal injury”). Supplementary searches addressed sleep interventions in athletes, potential physiological mechanisms, and evidence from military populations. All publications identified through Consensus were verified against their PubMed records, and information used in the review was obtained from the original abstracts or full texts rather than from AI-generated summaries.

Publications were selected purposively on the basis of their relevance to sleep quantity or quality and musculoskeletal injury in athletes. Prospective, retrospective, and cross-sectional observational studies and systematic reviews were treated as direct evidence. Mechanistic publications, expert opinions, sleep-intervention studies without injury outcomes, and studies in military populations were used only as contextual evidence. Studies addressing concussion exclusively were excluded from the core synthesis.

Because this was a narrative review, no preregistered protocol, duplicate independent screening, formal risk-of-bias assessment, or meta-analysis was performed. Consequently, the search should not be considered exhaustive, and selective retrieval of publications cannot be excluded.

2.3. Eligibility criteria

Publications were considered for inclusion if they involved athletic populations of any age, competitive level, or sport; reported a measure of sleep quantity, sleep quality, or a diagnosed sleep disorder; and reported an association, or the absence of one, with musculoskeletal injury, injury incidence, or injury risk. Systematic reviews addressing this relationship were included to summarize pooled or synthesized evidence. Studies addressing sport-related concussion exclusively, rather than musculoskeletal injury, were not treated as core evidence. Studies reporting only fatigue, delayed-onset muscle soreness, reaction time, physical performance, recovery markers, or subjective wellbeing, without a musculoskeletal injury outcome, were treated as supporting or mechanistic evidence rather than direct evidence of injury risk. Mechanistic and hypothesis-generating reports, narrative reviews, and evidence from closely related but non-athletic physically active populations (for example, military trainees) were included for context, clearly identified as such, and not presented as equivalent to primary athlete-injury evidence. Only English-language publications were considered for inclusion to the studies ultimately cited, and no further eligibility restrictions beyond those stated above (population, exposure, outcome, and evidentiary role) were applied.

2.4. Study selection and evidence use

Publications were selected purposively according to their direct relevance to sleep quantity or quality and musculoskeletal injury outcomes in athletic populations. Priority was given to prospective cohort studies and systematic reviews reporting clearly defined injury outcomes. Where full text was available, only information present in the full text was used. Where only an abstract was available, only information explicitly stated in that abstract was used, and no quantitative detail was inferred beyond what was reported. Studies drawn from the same underlying research project or overlapping cohorts were identified and are described as such rather than treated as fully independent sources of evidence.

2.5. Methodological limitations of this review

This narrative review has several limitations related to its methodology. First, as a narrative rather than systematic review, source selection was not governed by a pre-registered protocol, duplicate independent screening was not performed, and the search, while structured, cannot be assumed to have captured every relevant publication; some degree of selection bias cannot be excluded. Second, no formal meta-analysis or pooled effect estimate was calculated across the primary studies cited, given the substantial heterogeneity in sleep measurement, injury definition, follow-up duration, and population age described throughout the review. Third, Consensus was used only as a tool to assist in the identification of literature, and its outputs were not treated as a scientific source; all substantive claims are based on the primary

publications or systematic reviews themselves. Fourth, most of the primary evidence relies on self-reported sleep measures, which are subject to recall and reporting bias, while the small number of studies using objective actigraphy differ in follow-up duration and sample size, complicating direct comparison. Fifth, definitions of musculoskeletal injury vary considerably between studies, ranging from time-loss injury to any physical complaint, which limits the comparability of reported associations. Finally, English-language and full-text availability constraints may have limited access to some potentially relevant non-English or difficult-to-access publications.

3. REVIEW OF AVAILABLE EVIDENCE

3.1. Definitions and measurement of sleep quantity and quality

Sleep quantity is most commonly expressed as total sleep time, typically derived from self-report questionnaires or sleep diaries, or measured objectively with wrist actigraphy. Sleep quality is a broader construct that includes sleep efficiency (the proportion of time in bed spent asleep), sleep onset latency, the number and duration of awakenings after sleep onset, and subjective restfulness; these components are captured to varying degrees by instruments such as the Pittsburgh Sleep Quality Index, single-item Likert ratings of perceived sleep quality, and actigraphy-derived indices ¹. Chronic insomnia disorder is a distinct, clinically defined condition, characterized by persistent difficulty initiating or maintaining sleep with associated daytime impairment, and is diagnosed using structured clinical criteria rather than a simple duration or quality threshold ¹⁵. This distinction matters for interpreting the literature reviewed here: most athlete studies measure short sleep duration or reduced subjective sleep quality, not diagnosed insomnia, and the two exposures are not interchangeable.

Across the studies reviewed, the operational definition of "short" sleep duration varied, most commonly using a threshold of less than seven or less than eight hours per night, chosen to align with general population sleep recommendations rather than an athlete-specific evidence base for those particular cut-points. Similarly, sleep quality was operationalized inconsistently, ranging from validated questionnaires to single-item ratings collected as part of broader wellness-monitoring batteries. This heterogeneity in exposure definition, discussed further in section 3.10, is one of the principal limitations to direct comparison of findings across studies.

3.2. Sleep characteristics in athletic populations

Several reviews describe athletic populations as being at elevated risk of inadequate sleep relative to general population recommendations. Narrative reviews attribute this to a combination of sport-specific factors, including early-morning training sessions, travel across time zones, evening competition start times, and pre-competition anxiety, together with general risk factors that also apply to non-athletes, such as female sex, increasing age, and psychological stress or maladaptive perfectionism ¹. The consensus statement noted that retiring earlier before early-morning training did not fully offset the associated reduction in sleep duration. It also reported an association between sport type and chronotype distribution, with morning chronotypes more common in sports with early-morning training schedules ¹.

Reviews synthesizing objective and subjective sleep data in elite athletes report that a substantial proportion of athletes do not meet general sleep duration recommendations and that poor subjective sleep quality is common, although the pooled figures cited in these secondary sources originate from primary studies not independently reviewed here and are therefore presented cautiously, as background context rather than verified injury-outcome evidence ^{2,3}. Taken together, these background reviews support the plausibility that athletic populations experience a meaningful degree of sleep curtailment, which is the necessary premise for investigating whether this curtailment translates into elevated musculoskeletal injury risk.

3.3. Sleep duration and injury risk

The most frequently studied exposure in this literature is habitual sleep duration, generally dichotomized around a threshold of seven or eight hours per night.

In a cross-sectional study of 112 adolescent athletes at a combined middle and high school, self-reported habitual sleep duration was assessed at a single time point and compared against a retrospective review of injury records covering the preceding 21 months; this design does not constitute prospective follow-up of sleep exposure preceding subsequent injury, and should not be described as a prospective cohort. Athletes who reported sleeping fewer than eight hours per night were more likely to have had an injury recorded during the preceding 21 months than those reporting eight hours or more of sleep, both in univariate analysis and after multivariable adjustment; increasing school grade was an independent additional risk factor ⁶. A systematic review and meta-analysis, published first as a conference abstract and subsequently as a full peer-reviewed article, addressed adolescents aged 19 years or younger with sports or musculoskeletal injury more broadly, rather than a population restricted to defined athletic cohorts. The quantitative pooled estimate was derived from three heterogeneous studies, at least one of which included non-sport populations and non-sport injury mechanisms such as falls and bicycle-related trauma; this pooled estimate should therefore not be read as representative of seven independent athlete cohorts. With this caveat, most of the seven studies identified in the broader review individually reported associations between chronic short or poor sleep and increased injury risk in adolescents, while findings for acute, single-night sleep loss were less consistent ⁷.

In a cohort of 340 enrolled Swedish adolescent elite athletes, the prospective risk-factor analysis included 162 athletes who were uninjured at baseline. Sleeping more than eight hours on weekdays was associated with lower odds of a new injury (OR 0.39, 95% CI 0.16–0.99), alongside meeting the stated recommendation for fruit, vegetables, and fish intake ⁸. In a separate, partially overlapping cohort of 496 adolescent elite athletes from the same underlying research project followed prospectively over 52 weeks, a composite risk profile combining increased training load and intensity with decreased sleep volume was the strongest identified predictor of new injury in Cox regression analysis, ahead of any single variable considered in isolation ²⁰. Because both studies draw on overlapping recruitment from the same adolescent elite-athlete screening project, they are considered here as related rather than fully independent sources of evidence.

In adult endurance athletes, a 52-week prospective study of 95 participants from running, triathlon, swimming, cycling, and rowing disciplines found that sleeping fewer than seven hours per day was associated with significantly increased injury risk using a 14-day lag period, while sleeping more than seven hours per day was associated with reduced risk relative to the same reference category ¹³. By contrast, in a systematic review restricted to adult athletic populations, most individual studies addressing sleep duration, including several not directly reviewed here in full text, were judged to provide insufficient evidence of an independent association once study quality and consistency were taken into account, and the review's overall conclusion was that current evidence does not support poor sleep as an independent risk factor for injury in adult athletes ⁹. Evidence from non-athletic but physically demanding contexts provides some external context. In a large cohort of United States Army basic combat trainees, 45 candidate variables were considered and 27 were retained in the final total-cohort model, including PSQI-derived sleep variables; the analysis was not designed to isolate sleep as a standalone risk factor ¹⁶. A systematic review of military populations similarly reported that a majority of the small number of available studies found some association between sleep disturbance and musculoskeletal injury ¹⁷. This military evidence is not directly generalizable to competitive athletes and should be read as contextual rather than as confirmatory of an association between sleep duration and injury in athletic populations.

3.4. Sleep quality and injury risk

Sleep quality, distinct from sleep duration, has also been examined as a predictor of musculoskeletal injury, generally with a smaller number of available studies and greater reliance on either subjective ratings or objective sleep-efficiency measures from actigraphy.

In a nine-month prospective study of 17 female National Collegiate Athletic Association Division I volleyball players, who completed daily self-report ratings of sleep duration and sleep quality alongside mood, stress, and soreness, both decreased sleep duration and lower sleep quality were significant univariate predictors of subsequent time-loss injury; in models pairing sleep duration with each wellbeing measure, sleep duration remained significant, as did mood and soreness in their respective models; sleep quality was not included in these multivariable models ¹¹. In a prospective cohort of 23 elite Brazilian male soccer players who underwent ten nights of pre-season actigraphy monitoring followed by six months of injury surveillance through club medical records, lower sleep efficiency was associated with a greater number of injuries, greater injury severity, and more days of injury-related absence, with sleep efficiency accounting for a substantial proportion of the variance in each of these outcomes in linear regression ¹².

Not all studies using objective sleep-quality measures have reported this association. In a season-long study of 22 elite Australian Football League players, in which sleep duration and sleep efficiency were measured with actigraphy on the nights surrounding matches, no significant effect of sleep on injury occurrence was found when the five-night block surrounding an injury game was compared with the preceding or closest two non-injury weeks, and sleep parameters in the injury week were, if anything, slightly more favorable than in

preceding weeks rather than worse ¹⁰. This null finding should be interpreted cautiously given the small sample (n=22), single club and season, and limited number of injuries.

In a matched analysis of collegiate student-athletes, October sleep disturbance was associated with January injury status (OR 1.07 per PROMIS T-score unit, 95% CI 1.02–1.13); 41 of the 89 January cases were new, whereas 48 were already present in October ¹⁴. In the previously described Swedish HIFAA cohort of adolescent elite athletes, sleeping at least eight hours per night was associated with lower odds of injury within one year in multivariable analysis, but this whole-cohort estimate did not reach conventional statistical significance (OR 0.38, 95% CI 0.14-1.01). A borderline result was observed in the female subgroup only (OR 0.16, 95% CI 0.03-1.00, p=0.049); this exploratory subgroup finding should not be interpreted as established evidence of a sex-specific effect, and the corresponding non-significant result in male athletes should not be read as evidence of no effect, discussed further in section 3.7 ²¹.

Based on the abstract-level information available for this review, a 2026 scoping review included 17 studies involving athletes from 10 sports. Seven studies evaluated the relationship between actigraphy-derived sleep metrics and injury occurrence, and four of these seven (57%) reported that decreased sleep metrics correlated with increased injury occurrence. The review concluded that the evidence concerning injury risk remained inconclusive ²².

3.5. Evidence in youth athletes

The evidence base linking sleep to musculoskeletal injury points in a broadly similar direction more consistently in adolescent than in adult athletes, although the studies contributing to this pattern differ considerably in design and are not all specific to defined athlete cohorts. The adolescent evidence described above, drawn from school-based and elite-athlete populations in the United States and Sweden and from a broader systematic review and meta-analysis, generally points toward an association between shorter sleep and a higher occurrence of injury. However, the studies reported non-equivalent measures of association, including risk ratios, odds ratios, and hazard ratios; therefore, their numerical estimates should not be presented as a common range or compared directly. Moreover, the largest estimate concerned a composite profile combining increased training load and intensity with decreased sleep volume, rather than the independent effect of sleep alone ^{6,7,8,20,21}. This apparent convergence should not be read as an established, causally confirmed consensus; it summarizes directionally similar observational findings obtained with different designs and populations.

This relative consistency should be interpreted with several caveats. First, the U.S. school-based study contributing the 1.7-fold estimate was a cross-sectional sleep survey linked to a retrospective injury record review rather than a prospective cohort, which limits its evidentiary weight relative to the prospectively followed Swedish cohorts ⁶. Second, the pooled meta-analytic estimate from the broader adolescent review was derived from only three heterogeneous studies, at least one of which included non-sport populations and injury mechanisms, and the review's population was adolescents with sports or musculoskeletal injury generally rather than a population restricted to defined athletic cohorts; the individual studies it identified varied in design from cross-sectional to prospective, and in injury ascertainment

from self-report to structured monitoring ⁷. Third, two of the most frequently cited Swedish adolescent cohorts originate from the same underlying research project and cannot be treated as fully independent replications, as noted in section 3.3 ^{8,20}. Fourth, most adolescent studies rely on self-reported sleep duration rather than objective measurement, introducing the possibility of recall or reporting bias common to adolescent self-report research generally. Taking these caveats together, the observed association between short sleep and injury risk in adolescents is better supported than in adults, but it rests on a smaller and less independent body of evidence than a simple count of "three cohorts plus a meta-analysis" would suggest, and may partly reflect genuine developmental differences, such as the greater sleep needs and ongoing musculoskeletal development of adolescents, or differences in study design and publication patterns between the adolescent and adult literatures.

3.6. Evidence in adult, collegiate, and elite athletes

The most comprehensive synthesis of evidence in adult athletic populations is a systematic review that identified twelve eligible studies spanning professional and elite athletes, collegiate athletes, and dancers, assessed using the Newcastle-Ottawa Scale, with a mean quality score in the poor-to-fair range and only three of the twelve studies using objective rather than self-reported sleep measurement ⁹. That review concluded that, across professional and elite athletes, collegiate athletes, and dancers considered separately, current evidence was insufficient to support poor sleep as an independent risk factor for sport or training-related musculoskeletal injury, a conclusion that stands in contrast to the more consistent adolescent literature summarized in section 3.5 ⁹.

Individual adult and collegiate studies published within or after that review present a genuinely mixed picture rather than uniform absence of association. Among endurance athletes, sleep duration below seven hours per day was prospectively associated with increased injury risk ¹³. Among elite female collegiate volleyball players, both sleep duration and sleep quality predicted subsequent time-loss injury, with sleep duration remaining significant after adjustment for related wellbeing measures ¹¹. Among elite male soccer players, reduced pre-season sleep efficiency was associated with a greater number, severity, and duration of subsequent injuries ¹². Among a broader sample of collegiate student-athletes assessed longitudinally, greater October sleep disturbance was associated with January injury status in a matched analysis; only 41 of the 89 January cases were new ¹⁴. Conversely, in a small study using repeated objective actigraphic monitoring (five nights per week around matches across a competitive season) in elite team-sport athletes, no association between sleep and injury occurrence was detected in a small sample of 22 players with a limited number of injuries, with sleep parameters not differing meaningfully between injury and non-injury weeks ¹⁰.

This heterogeneity of findings within the adult and elite category, ranging from clear positive associations in several prospective cohorts to a null finding in a small actigraphy-based study and an overall systematic review conclusion of insufficient evidence, means that the true magnitude of any relationship between sleep and musculoskeletal injury in adult athletes cannot be reliably estimated from the available literature: the studies differ substantially in exposure definition, injury definition, and sample size, and no adequately powered meta-

analysis has been performed that would allow a summary effect size to be stated with confidence. It should not be assumed that the comparatively more consistent pattern observed in adolescents generalizes to adult, elite, or professional athletic populations.

3.7. Differences by sex, sport, and training context

Several studies report subgroup differences that raise the possibility, without formally demonstrating it, that the sleep-injury relationship may not operate uniformly across sexes or training contexts. In the Swedish HIFAA cohort, 171 adolescent elite athletes were enrolled and 125 were included in the multivariable analysis; female sex was an independent risk factor for injury within one year. The whole-cohort association between sleeping at least eight hours per night and reduced injury odds did not reach conventional statistical significance (OR 0.38, 95% CI 0.14-1.01); a borderline result was observed in the female subgroup (OR 0.16, 95% CI 0.03-1.00, $p=0.049$) and no significant result in the male subgroup. Because no formal statistical test for a sex-by-sleep interaction was reported, this sex-stratified pattern should be read as an exploratory subgroup finding that generates a hypothesis about a possible sex difference, rather than as demonstrated evidence that sex modifies the sleep-injury relationship. In the same cohort, high weekly training volume (nine or more sessions per week) was a significant risk factor overall and in the male subgroup specifically, but not in the female subgroup analyzed separately ²¹; again, no formal interaction test was reported, so this subgroup contrast should be treated with similar caution.

In the biopsychosocial analysis of 496 Swedish adolescent elite athletes across sixteen sports, the strongest identified risk factor was a three-component risk index combining increased training load, increased training intensity, and decreased sleep volume (HR 2.25, 95% CI 1.46–3.45); decreased sleep volume was also significant as a separate term in Model I (HR 1.42, 95% CI 1.06–1.89) ²⁰. Formal interactions were tested between gender and covariates and between significant covariates; no interactions were found. The systematic review of adult athletic populations reported that conclusions differed when professional and elite athletes, collegiate athletes, and dancers were considered as separate groups, but this reflects separate evidence synthesis within each subgroup rather than a direct, statistically tested comparison of sport type as an effect modifier ⁹. Taken together, these findings suggest that sex, training load, and sport type are plausible candidates for future, formally tested effect-modification analyses and independent replication; the evidence reviewed here does not establish that they modify the sleep-injury relationship, and sport type in particular should not be treated as a demonstrated modifier without direct interaction-test data.

3.8. Potential biological and behavioral mechanisms

Several mechanistic pathways have been proposed to explain how curtailed or poor-quality sleep could plausibly increase musculoskeletal injury risk, although it should be emphasized that most of this evidence is drawn from mechanistic and narrative reviews rather than from studies directly linking these intermediate pathways to confirmed musculoskeletal injury outcomes in athletes.

A hormonal and metabolic pathway has been proposed in which sleep curtailment shifts the balance between anabolic and catabolic signaling in skeletal muscle: slow-wave sleep is associated with growth hormone release important for tissue repair, while sleep restriction has been linked in experimental and narrative literature to reduced testosterone and growth hormone and increased cortisol, potentially favoring protein degradation over synthesis and compromising muscle tissue integrity and recovery capacity. This pathway is presented explicitly as a hypothesis in the literature rather than as an established causal mechanism confirmed in athletic injury cohorts ⁴.

A neurocognitive and neuromuscular pathway has also been proposed whereby sleep loss may impair attention, reaction time, working memory, and motor coordination. These waking functions are relevant to safe sport performance, but the available evidence does not establish that impairment of any one of them mediates musculoskeletal injury risk in athletes⁵. Sleep restriction may alter immune function and increase pro-inflammatory signalling. Whether these changes delay recovery from exercise-related microtrauma or increase subsequent musculoskeletal injury risk in athletes has not been established⁵. A broader neuroimmune framework, described in a recent review not specific to athletic populations, proposes bidirectional pathways linking poor sleep, psychological distress, and musculoskeletal pain and tissue health through shared inflammatory and central-sensitization mechanisms; this framework is presented as applicable to musculoskeletal health generally and has not been tested specifically as a predictor of new injury incidence in athletes ²³.

These mechanistic proposals provide biological plausibility for the associations described in sections 3.3 to 3.6, but plausibility alone does not establish that sleep curtailment is a direct cause of musculoskeletal injury in athletes; reverse causation, whereby pain, injury-related anxiety, or altered training schedules disrupt sleep rather than the reverse, and shared confounding by training load or psychological stress remain plausible alternative or contributing explanations, discussed further in section 4.

3.9. Sleep interventions and injury prevention

Relatively few studies have tested whether intervening on athletes' sleep reduces subsequent musculoskeletal injury; the majority of intervention research in this field has instead examined effects on athletic performance and recovery markers rather than injury incidence directly, and this distinction should be kept in mind when considering the applicability of this evidence to injury prevention.

A systematic review of ten intervention studies in athletes aged approximately 18 to 24 years identified sleep extension as the intervention most consistently associated with improved subsequent performance, while napping, sleep hygiene education, and post-exercise recovery strategies produced more mixed results across the included studies ²⁴. An updated systematic review identified twenty-five intervention studies published up to 2021, again concluding that sleep extension and napping had the most consistent supporting evidence, while sleep hygiene education alone, removal of electronic devices, and cold-water immersion showed limited or inconsistent effects, based on a still relatively small evidence base ²⁵. Neither of these

systematic reviews reported musculoskeletal injury as a primary outcome; their relevance here is in establishing that athletes' sleep is, to some degree, a modifiable target through behavioral intervention, which is a necessary precondition for any future injury-prevention trial targeting sleep.

A recent expert-opinion narrative review, explicitly identified by its authors as Level V evidence rather than primary research, has proposed that cognitive behavioral therapy for insomnia and mental-skills training, including mindfulness-based approaches, could plausibly reduce sports injury risk by improving sleep quality and perceptual-cognitive function relevant to injury avoidance; this proposal is presented as a rationale for future intervention research rather than as evidence of a demonstrated reduction in injury incidence ²⁶. An international expert consensus statement similarly recommends a structured approach to athlete sleep, including education, individualized screening, and behavioral strategies, as part of routine athlete health management, again as a general recommendation rather than as an injury-prevention trial with confirmed outcomes ¹. On the basis of the evidence reviewed here, no completed trial was identified, among the studies and reviews retrieved for this narrative review, in which a sleep-focused intervention was shown to directly and significantly reduce musculoskeletal injury incidence in athletes. Because the underlying literature search could not be fully documented or independently reconstructed (section 2.2), this statement should not be read as a claim that no such trial exists in the wider published literature; it remains an important question for future controlled research and, ideally, a dedicated systematic search.

3.10. Limitations and inconsistencies of the evidence

Several recurring methodological features limit the strength of conclusions that can be drawn from the evidence reviewed above. First, sleep exposure was measured inconsistently across studies: most relied on self-reported duration or quality, collected either as a single retrospective estimate, a periodic questionnaire, or a daily diary entry, while only a minority used objective actigraphy. Actigraphy-based evidence remains limited and inconsistent. In a 2026 scoping review, four of seven studies evaluating injury occurrence reported correlations between decreased sleep metrics and increased injury occurrence, while the review's overall conclusion was that evidence concerning injury risk remained inconclusive ^{10,12,22}. Differences between populations, sleep metrics, monitoring periods, and injury outcomes limit direct comparison.

Second, definitions of musculoskeletal injury varied substantially, from strict time-loss definitions, to medical-attention injuries, to any reported physical musculoskeletal complaint, complicating direct comparison of reported risk estimates across studies and making it inappropriate to combine them into a single pooled estimate within this narrative review ^{13,6,11}. Third, the systematic review specific to adult athletic populations explicitly rated the overall quality of the underlying primary studies as poor to fair using a validated quality-assessment tool, reflecting small sample sizes, limited adjustment for confounding, and reliance on self-report in most included studies ⁹; this quality profile applies, to varying degrees, across much of the adult literature reviewed in section 3.6.

Fourth, most primary studies reviewed here measured short sleep duration or reduced subjective sleep quality rather than clinically diagnosed sleep disorders; evidence linking newly diagnosed insomnia identified from administrative claims to musculoskeletal injury, reported in a large military cohort, cannot be assumed to generalize directly to the milder, subclinical sleep restriction more typical of the athlete studies described above, since insomnia disorder and short sleep duration are conceptually and diagnostically distinct exposures^{18,15}. Fifth, several of the studies with the strongest reported associations were prospective but observational, meaning that reverse causation, whereby pain or injury disrupts subsequent sleep rather than the reverse, cannot be excluded, particularly where sleep and injury were assessed close together in time¹⁴. Sixth, most reviewed studies did not adjust comprehensively for training load, psychological stress, or prior injury history simultaneously with sleep, so residual confounding by these related and plausibly correlated exposures remains a concern throughout this literature. Finally, much of the evidence for adolescent athletes derives from a small number of national or regional cohorts, including two studies drawn from the same underlying research project, which limits the independent replication of adolescent findings described in section 3.5.

Table 1. Characteristics of the key included studies.

Study (authors, year)	Study design / population / sport	Sample size	Sleep assessment	Injury definition / follow-up	Main findings	Main limitations
Milewski et al, 2014	Cross-sectional sleep survey with retrospective injury record review (not a prospective cohort); Adolescent, school-based (grades 7-12); Multiple sports	n=112 (54M/58F), mean age 15.2y	Self-reported average nightly sleep duration	Sports injuries documented in school athletic-training records during the preceding 21 months; retrospective look-back period, not prospective follow-up	<8h/night; RR 1.7 (95% CI 1.0-3.0) multivariable; increasing grade also independent risk factor	Single-school study; sleep assessed once after the injury-record period; injuries were obtained from retrospective records rather than participant recall; temporal ordering and causal direction cannot be established
Gao et al, 2019 (systematic review and meta-analysis)	Systematic review and meta-analysis, 7 studies identified; Adolescents (<=19y) with sports or musculoskeletal injury; not restricted to defined athlete cohorts; Multiple sports and some non-sport settings (pooled)	7 studies identified; quantitative pooled estimate based on only 3 heterogeneous studies	Varied (self-report across included studies)	Varied, including some non-sport injury mechanisms (e.g., falls, bicycle-related trauma); follow-up: Varied	Chronic short/poor sleep: pooled OR 1.58 (95% CI 1.05-2.37) from 3 heterogeneous studies; acute sleep loss findings mixed	Pooled estimate not specific to athletes and based on only 3 heterogeneous studies; substantial between-study heterogeneity in population, exposure, and outcome definitions
von Rosen et al, 2016 (2017 print)	Two-wave prospective cohort study, two survey time points; Adolescent elite athletes (KASIP project); 7 sports	n=340 enrolled (178M/162F); n=162 in the prospective risk-factor analysis	Self-reported sleep questionnaire	Self-reported new injury; follow-up: Two semesters (autumn/spring)	Sleeping >8 h on weekdays at baseline was associated with lower odds of a new injury at follow-up (OR 0.39, 95% CI 0.16–0.99)	Self-reported exposure and outcome; only two assessment occasions; limited adjustment for confounding; possible attrition and overlapping recruitment with the other KASIP cohort; causality cannot be inferred.
von Rosen et al, 2017	Prospective cohort; Adolescent elite athletes (KASIP project, overlapping with Row 3); 16 sports	n=496, age 15-19y	Repeated self-reported sleep volume, assessed weekly or biweekly depending on sport.	New injury (weekly monitoring); follow-up: 52 weeks	The composite profile of increased training load and intensity together with decreased sleep volume was the strongest identified risk profile in the multivariable Cox	Composite, time-varying exposure; decreased sleep volume was also significant as a separate model term; interaction testing found no interactions; sleep and injury were self-reported;

Study (authors, year)	Study design / population / sport	Sample size	Sleep assessment	Injury definition / follow-up	Main findings	Main limitations
					model (HR 2.25, 95% CI 1.46–3.45).	recruitment overlapped with the related KASIP study.
Haraldsdottir et al, 2021	Prospective cohort, daily self-report; Female NCAA Division I collegiate athletes; Volleyball	n=17	Daily self-reported sleep duration and quality (Likert)	Time-loss injury; follow-up: 9 months	Sleep duration OR 0.69 (95% CI 0.55-0.87); sleep quality OR 0.49 (95% CI 0.34-0.70); duration remained independent predictor after adjustment	Very small single-team sample; single sport, single sex
Dennis et al, 2016	Prospective cohort, actigraphy; Elite professional team-sport athletes; Australian Football League	n=22	Actigraphy (5 nights/week around matches)	Game-related time-loss injury; follow-up: One competitive season (2013)	No significant association between sleep duration/efficiency and injury occurrence (null finding)	Small sample; single club/season; objective measure but limited statistical power
Johnston et al, 2020	Prospective cohort, weekly diary; Adult endurance athletes; Running, triathlon, swimming, cycling, rowing	n=95, mean age 42y	Self-reported weekly sleep quantity	Any physical musculoskeletal complaint/impairment; follow-up: 52 weeks	<7h/day sleep (14-day lag): HR 1.51 (95% CI 1.13-2.02); >7h/day: HR 0.63 (95% CI 0.45-0.87)	Self-reported sleep and injury; heterogeneous endurance disciplines pooled
Dobrosielski et al, 2021 (systematic review)	Systematic review, 12 studies, NOS quality-assessed; Adult athletic populations; Professional/elite, collegiate, dancers (across included studies)	12 studies (pooled populations)	Varied (mostly self-report; 3/12 objective)	Varied across included studies; follow-up: Varied	Overall: insufficient evidence for sleep as an independent injury risk factor in adult athletes	Mean quality score poor-to-fair (NOS); high heterogeneity precluded meta-analysis
Silva et al, 2020	Prospective cohort; Elite male professional athletes; Soccer	n=23	Actigraphy (10 nights, pre-season)	Injury count, severity, and absence days (club medical records); follow-up: 6 months	Lower sleep efficiency associated with more injuries, greater severity, more absence days (linear regression)	Sleep measured only during a 10-day pre-season window, not throughout follow-up; small sample
Messman et al, 2025	Matched case-control, longitudinal; Collegiate student-athletes (NCAA, multiple divisions); Multiple sports	n=89 matched pairs	PROMIS Sleep Disturbance T-score (self-report)	Self-reported January injury status (41 new cases and 48 cases persisting from October); follow-up: 3 months (two time points, Oct 2020/Jan 2021)	Sleep disturbance OR 1.07 per unit increase (p=0.007); injury probability increased with disturbance severity category	Predominantly female/White sample; COVID-disrupted competition context; self-reported injury
Abrahams et al., 2025 (HIFAA)	Prospective cohort; Adolescent elite athletes (HIFAA project); 8 sports	n=171 enrolled (104M/67F), mean age 16.9y; n=125 in the multivariable analysis	Self-reported habitual sleep duration (baseline questionnaire)	Self-reported new injury within 1 year; follow-up: 1 year	Sleep $\geq$ 8h/night: OR 0.38 (95% CI 0.14-1.01) overall; OR 0.16 (95% CI 0.03-1.00) in females only; female sex OR 3.51 independent risk factor; high training volume OR 3.98 in males	Missing data in ~25/171 for injury outcome; sex-stratified subgroup analyses have limited power
Lytle et al., 2026	Scoping review; athletes from 10 sports	17 included studies; 7 studies evaluated injury occurrence	Actigraphy	Injury occurrence	4 of 7 injury studies reported correlations between decreased sleep metrics and increased injury occurrence; overall evidence concerning injury risk was considered inconclusive	Only abstract-level information was available for the present narrative review; detailed study-level appraisal was not possible

Abbreviations: CI, confidence interval; HR, hazard ratio; KASIP, Karolinska Athlete Screening Injury Prevention (project); NCAA, National Collegiate Athletic Association; NOS, Newcastle-Ottawa Scale; OR, odds ratio; PROMIS, Patient-Reported Outcomes Measurement Information System; RR, relative/risk ratio.

Table 2. Summary of evidence by sleep exposure and injury outcome.

Sleep exposure	Population	Summary of findings and main limitations	Supporting sources
Short sleep duration (self-reported, typically <7-8h/night)	Adolescents	Association reported	Milewski 2014; Gao 2019 (SR/MA); von Rosen 2016
Short sleep duration (self-reported)	Adults/collegiate	Mixed: association reported in some cohorts, not confirmed as independent factor in systematic review	Johnston 2020 (association); Dobrosielski 2021 (SR, insufficient evidence overall)
Composite training-load-and-sleep profile	Adolescents	Association reported for a composite profile; decreased sleep volume was also significant as a separate model term; interaction testing found no interactions.	von Rosen 2017
Self-reported sleep duration	Adolescents	Whole-cohort estimate not statistically significant (OR 0.38, 95% CI 0.14-1.01); borderline, exploratory association in female subgroup only (OR 0.16, p=0.049)	Abrahamson et al., 2025 (HIFAA)
Self-reported sleep quality or sleep disturbance	Adults/collegiate	Association reported	Haraldsdottir 2021; Messman 2025
Objective sleep efficiency (actigraphy)	Adults/elite	Association reported (single pre-season measurement)	Silva 2020
Objective sleep duration/efficiency (repeated actigraphy, 5 nights/week)	Adults/elite	No significant association (null finding); small sample (n=22), single club/season	Dennis 2016
Newly diagnosed insomnia identified from administrative claims	Non-athletic, active young adults (military)	Association reported (contextual evidence, not athlete population)	Wickwire 2025
Self-reported sleep quality/duration	Non-athletic, active young adults (military)	Sleep contributed within a multivariable risk-prediction model (Foulis, not an isolated exposure); majority of a small evidence base reported an association (Lisman, systematic review)	Foulis 2025; Lisman 2022 (SR)

Note: "Association reported" indicates a statistically significant association was reported in at least one cited study; it does not imply consistent replication across all cited sources for that row.

Table 3. Main methodological limitations and research gaps.

Limitation / research gap	Description
Inconsistent sleep measurement	Objective studies remain few and heterogeneous. In a 2026 scoping review, four of seven studies evaluating injury occurrence reported correlations between decreased sleep metrics and increased injury occurrence; the overall evidence concerning injury risk was considered inconclusive ²²
Inconsistent injury definitions	Definitions range from strict time-loss injury to any physical musculoskeletal complaint, limiting comparability and precluding pooled quantitative synthesis
Small sample sizes	Several key studies included fewer than 30 participants (e.g., Dennis 2016, n=22; Silva 2020, n=23; Haraldsdottir 2021, n=17), limiting statistical power and precision of effect estimates
Overlapping cohorts	Two frequently cited adolescent studies (von Rosen 2016, von Rosen 2017) draw on the same underlying research project, limiting independent replication
Limited adjustment for confounding	Few studies adjusted simultaneously for training load, psychological stress, and prior injury history alongside sleep

Limitation / research gap	Description
Reverse causation not excluded	Observational designs, particularly with short intervals between sleep and injury assessment, cannot rule out injury-related pain or anxiety as a cause of poor sleep rather than the reverse
Age-related inconsistency unexplained	Evidence is more consistent in adolescents than in adults; whether this reflects true biological differences or differences in study quality and design is unresolved
Absence of injury-outcome intervention trials	Among the studies and reviews retrieved for this narrative review, no completed trial tested whether a sleep-focused intervention reduces musculoskeletal injury incidence; existing intervention research targets performance and recovery outcomes. Because the underlying search could not be fully reconstructed, this should not be read as a claim about the entire published literature
Limited generalisability	Evidence concentrated in a small number of countries, sports, and research groups; contextual evidence from military and non-athletic populations cannot be assumed to generalize directly to athletes
Search reproducibility not fully documented	Complete PubMed strategy, Consensus queries, filters, and record counts were not preserved in a fully reconstructable form; the completeness of the underlying literature search cannot be independently verified, and statements about the absence of certain study types are limited to the studies and reviews actually identified

4. DISCUSSION

This narrative review summarized evidence on the association of sleep quantity and quality with musculoskeletal injury risk in athletes. The findings were not consistent across all populations. In adolescent athletes, several data sources and a broader systematic review and meta-analysis more consistently indicated an association. However, some of the adolescent cohorts overlapped, and the population included in the meta-analysis was not restricted to clearly defined athlete populations. Findings in adult, collegiate, and elite athletes were inconsistent. A dedicated systematic review concluded that current evidence did not support sleep as an independent risk factor in adult athletic populations, whereas several individual studies reported significant associations and a small prospective actigraphy study involving 22 players from one club and one competitive season reported no statistically significant association^{6,7,8,9,10,11,12,13,14,20}.

Several factors plausibly contribute to this apparent age-related difference. Adolescents are still undergoing musculoskeletal growth and maturation, have generally higher physiological sleep requirements than adults, and may be less able to compensate behaviorally for sleep debt through strategies such as napping or professionalized recovery support that are more available to elite adult athletes. It is equally possible that the adult literature is simply noisier due to smaller, more heterogeneous samples, rather than reflecting a true absence of association; the systematic review of adult athletes explicitly rated most included studies as poor to fair in methodological quality, which limits confidence in a null conclusion as much as it would a positive one⁹. Actigraphy-based evidence is also heterogeneous. In a 2026 scoping review, four of seven studies evaluating injury occurrence reported correlations between decreased sleep metrics and increased injury occurrence. However, the review concluded that the overall evidence concerning injury risk remained inconclusive²². These findings do not support either a uniformly positive or a uniformly null interpretation of objectively measured sleep.

The role of confounding and reverse causation deserves particular caution in this literature. Training load, competition schedule, and psychological stress are plausible common causes of both poor sleep and injury risk, and few of the reviewed studies adjusted comprehensively for

these factors alongside sleep. Injury itself, and the pain, anxiety, and disrupted routine that frequently accompany it, may independently cause subsequent poor sleep, raising the possibility of reverse causation in observational designs, particularly where sleep and injury were measured close together in time ¹⁴. The proposed hormonal, neurocognitive, and immune-inflammatory mechanisms linking sleep to tissue repair and injury susceptibility offer biological plausibility for a causal pathway from sleep to injury, but these mechanisms have not been directly tested as mediators in athlete injury cohorts, and a neuroimmune framework linking sleep, psychological distress, and musculoskeletal health explicitly describes bidirectional rather than strictly unidirectional pathways ^{4,5,23}. On the basis of the evidence reviewed here, it would not be appropriate to conclude that sleep curtailment causes musculoskeletal injury in athletes; the more defensible conclusion is that short sleep duration and poor sleep quality are associated with increased injury risk in several, though not all, athletic populations studied, most consistently in adolescents.

The exploratory subgroup differences and co-occurring risk profiles observed by sex and training load, seen most clearly in the Swedish adolescent HIFAA cohort and in the biopsychosocial analysis of Swedish adolescent elite athletes, raise the possibility, without formally demonstrating it, that a single population-wide sleep-injury relationship may not apply uniformly ^{20,21}. These subgroup findings arise from relatively modest sample sizes and should be regarded as hypothesis-generating rather than confirmatory. HIFAA did not report a sex-by-sleep interaction test, whereas von Rosen et al. reported no significant interactions; future studies should use adequately powered interaction analyses to test effect modification by sex, sport type, and concurrent training load.

The generalizability of this evidence base is limited by its geographic and demographic concentration. Several of the most frequently cited studies originate from a small number of countries and research groups, and the adult evidence in particular is drawn from a modest number of small cohorts across different sports, which limits confidence that findings from one sport or competitive level generalize to others ⁹. Evidence from newly diagnosed insomnia identified from administrative claims in military populations, while methodologically strong in terms of sample size, involves a different exposure, a different population, and a different injury-ascertainment system than the athlete cohorts reviewed here, and should be regarded as contextual rather than directly generalizable evidence ^{18,16,17}.

The available evidence does not support a general claim that improving sleep reduces musculoskeletal injury risk in all athletic populations. No intervention trial with injury as a primary outcome was identified among the studies and reviews retrieved for this narrative review. Because the literature search was incompletely documented, this finding should not be interpreted as evidence that no such trials exist in the wider literature. Supporting adequate sleep remains reasonable for general athlete health; however, current evidence does not establish sleep-focused care as an effective musculoskeletal injury-prevention intervention. This position is supported by the plausible mechanistic rationale, the relative safety and low cost of sleep-focused education, and the more consistent evidence in adolescents, but it requires confirmation in intervention studies. Future prospective studies should combine objective and subjective sleep measurements, use standardized, consensus-based injury definitions, adjust

simultaneously for training load and psychological stress, and test effect modification by sex, age, and sport. Such studies are needed to determine which athletes, if any, are most likely to benefit from sleep-focused injury-prevention strategies. The published HIFAA protocol and subsequent cohort findings illustrate the type of prospective, systematically monitored design needed to address the remaining uncertainties ²⁷.

5. CONCLUSIONS

The strength of the evidence differs between populations and sleep measures. The most consistent finding is an association between habitually short sleep duration and increased musculoskeletal injury risk in adolescent athletes; evidence for poor subjective sleep quality is less consistent. This conclusion is supported by findings in the same direction from several cross-sectional and prospective data sources and a broader systematic review and meta-analysis, although the data sources were not all independent and were not all restricted to clearly defined athlete cohorts. Associations have been reported in some adult, collegiate, and elite athletic populations, but findings remain inconsistent. Several prospective studies in these groups reported significant associations involving sleep duration, sleep quality, or sleep disturbance, whereas a dedicated systematic review of adult athletes concluded that the available evidence was insufficient to establish sleep as an independent risk factor in the broader adult athletic population.

It remains uncertain whether objectively measured sleep shows a consistent relationship with injury risk. Although four of seven injury studies included in a recent scoping review reported correlations involving poorer sleep metrics, the review concluded that the overall evidence concerning injury risk remained inconclusive ²². It also remains uncertain whether and how sex, sport type, and concurrent training load are associated with differences in any sleep-injury relationship; the available subgroup analyses are exploratory and have not been confirmed with formal interaction testing or independent replication. Finally, the specific biological mechanisms proposed to link sleep to musculoskeletal tissue health, including hormonal, neurocognitive, and immune-inflammatory pathways, remain plausible but unconfirmed as direct mediators of injury risk in athletes.

On the basis of the evidence currently available, it cannot be recommended that sleep-focused interventions be routinely implemented as a stand-alone musculoskeletal injury-prevention strategy in athletes: no completed trial demonstrating a reduction in injury incidence as a direct result of a sleep intervention was identified among the studies and reviews retrieved for this narrative review, though the incompletely documented literature search (section 2.2) means this cannot be generalized to a claim about the entire published literature. Supporting adequate sleep remains reasonable for general athlete health; however, current evidence does not establish sleep-focused care as an effective musculoskeletal injury-prevention intervention. Confirmatory prospective and intervention studies using standardized injury definitions and reproducible sleep-measurement methods are needed.

Disclosure

Supplementary Materials

Not applicable.

Author Contributions

Conceptualization, P.K., B.G., J.Li.; methodology, P.K., A.S., J.Pa.; investigation, P.K., L.R., O.S.; data curation, P.K., D.K.-O., J.Pr.; writing – original draft preparation, P.K., N.W., M.B.; writing – review and editing, P.K., G.K., J.La.; supervision, P.K., M.W. All authors have read and agreed to the published version of the manuscript.

Funding

The preparation of this manuscript received no external funding.

Institutional Review Board Statement

Not applicable.

Informed Consent Statement

Not applicable.

Data Availability Statement

All data discussed in this narrative review are available in the publications cited in the manuscript.

Acknowledgments

Not applicable.

Conflicts of Interest

The authors declare that they have no financial or non-financial conflicts of interest relevant to the content of this manuscript.

Originality

This manuscript is an original work and has not been published previously, in whole or in part, and is not currently under consideration for publication elsewhere. All authors have read and approved the submitted version.

Declaration of the use of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, Consensus was used as a supplementary tool to identify potentially relevant literature, and ChatGPT (OpenAI) was used for editorial assistance, including improvement of language, readability, organization, and formatting. All references were independently verified against PubMed and the original publications. The authors critically reviewed and revised the entire manuscript, retained full control over the interpretation of the evidence and the conclusions.

References

1. Walsh NP, Halson SL, Sargent C, Roach GD, Nédélec M, Gupta L, Leeder J, Fullagar HH, Coutts AJ, Edwards BJ, Pullinger SA, Robertson CM, Burniston JG, Lastella M, Le Meur Y, Hausswirth C, Bender AM, Grandner MA, Samuels CH. Sleep and the athlete: narrative review and 2021 expert consensus recommendations. *Br J Sports Med.* 2021;55(7):356-368. <https://doi.org/10.1136/bjsports-2020-102025>
2. Charest J, Grandner MA. Sleep and Athletic Performance: Impacts on Physical Performance, Mental Performance, Injury Risk and Recovery, and Mental Health. *Sleep Med Clin.* 2020;15(1):41-57. <https://doi.org/10.1016/j.jsmc.2019.11.005>
3. Simpson NS, Gibbs EL, Matheson GO. Optimizing sleep to maximize performance: implications and recommendations for elite athletes. *Scand J Med Sci Sports.* 2017;27(3):266-274. <https://doi.org/10.1111/sms.12703>
4. de Sousa Nogueira Freitas L, Rodrigues da Silva FR, de Araújo Andrade H, Guerreiro RC, Viegas Paulo FV, de Mello MT, Silva A. Sleep debt induces skeletal muscle injuries in athletes: A promising hypothesis. *Med Hypotheses.* 2020;142:109836. <https://doi.org/10.1016/j.mehy.2020.109836>
5. Kaczmarek F, Bartkowiak-Wieczorek J, Matecka M, Jenczylik K, Brzezińska K, Gajniak P, Marchwiak S, Kaczmarek K, Nowak M, Kmiecik M, Stężycka J, Krupa KK, Mądry E. Sleep and Athletic Performance: A Multidimensional Review of Physiological and Molecular Mechanisms. *J Clin Med.* 2025;14(21):7606. <https://doi.org/10.3390/jcm14217606>
6. Milewski MD, Skaggs DL, Bishop GA, Pace JL, Ibrahim DA, Wren TA, Barzdukas A. Chronic lack of sleep is associated with increased sports injuries in adolescent athletes. *J Pediatr Orthop.* 2014;34(2):129-133. <https://doi.org/10.1097/BPO.0000000000000151>
7. Gao B, Dwivedi S, Milewski MD, Cruz AI Jr. Lack of Sleep and Sports Injuries in Adolescents: A Systematic Review and Meta-analysis. *J Pediatr Orthop.* 2019;39(5):e324-e333. <https://doi.org/10.1097/BPO.0000000000001306>
8. von Rosen P, Frohm A, Kottorp A, Fridén C, Heijne A. Too little sleep and an unhealthy diet could increase the risk of sustaining a new injury in adolescent elite athletes. *Scand J Med Sci Sports.* 2017;27(11):1364-1371. <https://doi.org/10.1111/sms.12735>
9. Dobrosielski DA, Sweeney L, Lisman PJ. The Association Between Poor Sleep and the Incidence of Sport and Physical Training-Related Injuries in Adult Athletic Populations: A Systematic Review. *Sports Med.* 2021;51(4):777-793. <https://doi.org/10.1007/s40279-020-01416-3>

10. Dennis J, Dawson B, Heasman J, Rogalski B, Robey E. Sleep patterns and injury occurrence in elite Australian footballers. *J Sci Med Sport*. 2016;19(2):113-116. <https://doi.org/10.1016/j.jsams.2015.02.003>
11. Haraldsdottir K, Sanfilippo J, McKay L, Watson AM. Decreased Sleep and Subjective Well-being as Independent Predictors of Injury in Female Collegiate Volleyball Players. *Orthop J Sports Med*. 2021;9(9):23259671211029285. <https://doi.org/10.1177/23259671211029285>
12. Silva A, Narciso FV, Soalheiro I, Viegas F, Freitas LSN, Lima A, Leite BA, Aleixo HC, Duffield R, de Mello MT. Poor Sleep Quality's Association With Soccer Injuries: Preliminary Data. *Int J Sports Physiol Perform*. 2020;15(5):671-676. <https://doi.org/10.1123/ijsp.2019-0185>
13. Johnston R, Cahalan R, Bonnett L, Maguire M, Glasgow P, Madigan S, O'Sullivan K, Comyns T. General health complaints and sleep associated with new injury within an endurance sporting population: A prospective study. *J Sci Med Sport*. 2020;23(3):252-257. <https://doi.org/10.1016/j.jsams.2019.10.013>
14. Messman BA, Petrie KA, Moore EWG, Petrie TA. Sleep Disturbances and Risk of Sports Injury Among Collegiate Student-Athletes. *Clin J Sport Med*. 2025;35(5):e54-e60. <https://doi.org/10.1097/JSM.0000000000001278>
15. Mysliwiec V, Martin JL, Ulmer CS, Chowdhuri S, Brock MS, Spevak C, Sall J. The Management of Chronic Insomnia Disorder and Obstructive Sleep Apnea: Synopsis of the 2019 U.S. Department of Veterans Affairs and U.S. Department of Defense Clinical Practice Guidelines. *Ann Intern Med*. 2020;172(5):325-336. <https://doi.org/10.7326/M19-3575>
16. Foulis SA, Proctor SP, Spiering BA, Walker LA, Aaron KG, Castellani CM, Hussian IM, Heaton KJ, Bouxsein ML, Gaffney-Stomberg E, Fraley AL, Popp KL, Davis IS, Staab JS, Staab JE, Judkins JL, Merkle SL, Ritland BM, Matheny RW Jr, Hauret KG, Canham-Chervak M, McClung JP, Jones BH, Friedl KE, Hughes JM, Taylor KM; ARIEM Reduction in Musculoskeletal Injury (ARMI) Study Team. Model for Musculoskeletal Injury Risk Factors Among US Army Basic Combat Trainees. *JAMA Netw Open*. 2025;8(6):e2513177. <https://doi.org/10.1001/jamanetworkopen.2025.13177>
17. Lisman P, Ritland BM, Burke TM, Sweeney L, Dobrosielski DA. The Association Between Sleep and Musculoskeletal Injuries in Military Personnel: A Systematic Review. *Mil Med*. 2022;187(11-12):1318-1329. <https://doi.org/10.1093/milmed/usac118>
18. Wickwire EM, Capaldi VF, Herrin J, Stryckman B, Thomas C, Williams SG, Werner JK, Funk W, Nassif TH, Albrecht JS. Burden of Insomnia Disorder Among US Active-Duty Military Personnel. *Neurology*. 2025;105(9):e214106. <https://doi.org/10.1212/WNL.00000000000214106>
19. Baethge C, Goldbeck-Wood S, Mertens S. SANRA—a scale for the quality assessment of narrative review articles. *Res Integr Peer Rev*. 2019;4:5. <https://doi.org/10.1186/s41073-019-0064-8>
20. von Rosen P, Frohm A, Kottorp A, Fridén C, Heijne A. Multiple factors explain injury risk in adolescent elite athletes: Applying a biopsychosocial perspective. *Scand J Med Sci Sports*. 2017;27(12):2059-2069. <https://doi.org/10.1111/sms.12855>
21. Abrahamson J, Sandstedt E, Sansone M, Lindman I. Female sex, training volume and sleep amount are risk factors for sports-related injuries: a 1-year prospective cohort study

- on adolescent elite athletes. *BMJ Open Sport Exerc Med*. 2025;11(4):e002699. <https://doi.org/10.1136/bmjsem-2025-002699>
22. Lytle C, Krishnan P, Saraf SM, Mulcahey MK. Association between sports injuries and sleep as measured by sleep actigraphy: a scoping review. *Phys Sportsmed*. 2026;54(3):173-179. <https://doi.org/10.1080/00913847.2026.2626259>
 23. Klyne DM, Hall M, Smith SS, Barbe MF, Egorova-Brumley N, Serafimovska A, Hodges PW, Suraev A. Understanding the relationship between sleep, psychological and musculoskeletal health from a neuroimmune perspective. *Musculoskelet Sci Pract*. 2026;82:103509. <https://doi.org/10.1016/j.msksp.2026.103509>
 24. Bonnar D, Bartel K, Kakoschke N, Lang C. Sleep Interventions Designed to Improve Athletic Performance and Recovery: A Systematic Review of Current Approaches. *Sports Med*. 2018;48(3):683-703. <https://doi.org/10.1007/s40279-017-0832-x>
 25. Cunha LA, Costa JA, Marques EA, Brito J, Lastella M, Figueiredo P. The Impact of Sleep Interventions on Athletic Performance: A Systematic Review. *Sports Med Open*. 2023;9(1):58. <https://doi.org/10.1186/s40798-023-00599-z>
 26. Haskell B, Eiler A, Essien H. Sleep Quality and Cognitive Skills Impact Neurocognitive Function and Reduce Sports-Related Injury Risk. *Arthrosc Sports Med Rehabil*. 2025;7(2):101077. <https://doi.org/10.1016/j.asmr.2025.101077>
 27. Lindman I, Ryman Augustsson S, Baranto A, Sansone M, Abrahamson J. Prospective cohort study identifying risk factors for sports injuries in adolescent athletes: the Healthy Injury-Free Adolescent Athletes (HIFAA) study. *BMJ Open Sport Exerc Med*. 2024;10(3):e002253. <https://doi.org/10.1136/bmjsem-2024-002253>